1111 labmeeting

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1 previous

연구의 목표 살짝 변경.
추가적인 정보를 더 주는 건 맞는데, 그 정보의 data type 이 continuous 하기 때문에 정보량 차이가 발생하는 문제를 해결해야 함.
결론적으로는 이 과정을 통해서 LSM 의 고차 구조 복원을 더 잘 해보도록 하겠다는 것.

2 model

2.1 main model

$$g_{(\ell)}\left(Pr(Y_{ij} = y_{ij})\right) = \alpha^{(\ell)} + \theta_i^{(\ell)} + \theta_j^{(\ell)} - \beta^{(\ell)} d(u_i^{(\ell)}, u_j^{(\ell)})$$
$$u_i^{(\ell)} | z_i \sim N\left(z_i, I \cdot \sigma^2_{(\ell)}\right)$$

2.2 prior distribution

$$\alpha^{(\ell)} \sim \mathcal{N}(0,1)$$
$$\theta_i^{(\ell)} \sim \mathcal{N}(0,1)$$

해당 변수들은 일단 이렇게 뒀는데, Sosa (2022) 에서는 모든 변수에 대해 계층적인 구조를 추가했다. (0,1 대신 normal, invgamma prior 를 부여한 변수 사용했다는 것.) 이는 모델의 유연성을 추가하긴 하지만, karate network 만을 사용하는 현재 상황에서는 적당히 hyperparameter setting 해서 해결할 수 있을 것으로 봐서 일단 skip 했음.

$$\beta^{(\ell)} \sim \mathsf{Gamma}(a_0, b_0)$$

해당 변수는 non-negative 값을 가질 수 있도록 한다. Sosa (2022) 에서는 $exp(\theta)$ 값으로 nonegativity 를 지킴. 일단 딱히 바꿔야 할 이유를 찾지 못했고, exponential 은 기하급수적으로 커지는 성질 때문에 특정 layer 의 latent position 의 영향이 과대추정될까봐 일단 놔뒀음.

$$z_i \sim \mathcal{N}(0, 1)$$

로 고정하였음. latent position 의 경우 지나치게 유연하면 identifiability issue 가 더 심해질 것 같아서,,, 마지막으로 variance 관련 parameter 는 아래와 같이 정의했음.

$$\sigma^2_{(\ell)} \sim \mathsf{IG}(a_1, b_1)$$

2.3 update rule

let $\Theta_{ij}^{(\ell)} = \{\alpha^{(\ell)}, \theta_i^{(\ell)}, \theta_j^{(\ell)}, \beta^{(\ell)}, u_i^{(\ell)}, u_j^{(\ell)}\}$

2.3.1 update $\alpha^{(\ell)}$

with likelihood function:

$$\ell^{(\ell)}(\alpha^{(\ell)}|\Theta^{(\ell)}\backslash\alpha^{(\ell)}) = \prod_{i<j} \ell^{(\ell)}(\alpha^{(\ell)}|\Theta_{ij}^{(\ell)}\backslash\alpha^{(\ell)}) = \prod_{i<j} P_{(\ell)}\left(g_{(\ell)}^{-1}\left(\alpha^{(\ell)} + \theta_i^{(\ell)} + \theta_j^{(\ell)} - \beta^{(\ell)} d(u_i^{(\ell)}, u_j^{(\ell)})\right)\right)$$

therefore, with symmetric proposal distribution - Normal - , MH ratio is,

$$\text{MH ratio} = \frac{P(\alpha^{(\ell)*})\ell^{(\ell)}(\alpha^{(\ell)*}|\Theta^{(\ell)}\backslash\alpha^{(\ell)})}{P(\alpha^{(\ell)})\ell^{(\ell)}(\alpha^{(\ell)}|\Theta^{(\ell)}\backslash\alpha^{(\ell)})}$$

such that $\alpha^{(\ell)*}$ is proposed parameter.

2.3.2 update $\theta_i^{(\ell)}$

with likelihood function:

$$\ell^{(\ell)}(\theta_i^{(\ell)}|\Theta^{(\ell)}\backslash\theta_i^{(\ell)}) = \prod_{i<j} P_{(\ell)}\left(g_{(\ell)}^{-1}\left(\alpha^{(\ell)} + \theta_i^{(\ell)} + \theta_j^{(\ell)} - \beta^{(\ell)} d(u_i^{(\ell)}, u_j^{(\ell)})\right)\right)$$

therefore, with symmetric prior distribution - Normal - , MH ratio is,

$$\text{MH ratio} = \frac{P(\theta_i^{(\ell)*})\ell^{(\ell)}(\theta_i^{(\ell)*}|\Theta^{(\ell)}\backslash\theta_i^{(\ell)*})}{P(\theta_i^{(\ell)})\ell^{(\ell)}(\theta_i^{(\ell)}|\Theta^{(\ell)}\backslash\theta_i^{(\ell)})}$$

such that $\theta_i^{(\ell)*}$ is proposed parameter.

2.3.3 update $\beta^{(\ell)}$

with likelihood function same as above, and using normal distribuiton as proposal distribution, we have to use jacobian due to parameter transition for nonnegative restriction.
let $\log \beta^{(\ell)} = \kappa^{(\ell)}$, therefore jacobian is $\beta = \frac{d\beta^{(\ell)}}{d\kappa^{(\ell)}}$. So the MH ratio is defined as follow,

$$\text{MH ratio} = \frac{P(\beta^{(\ell)*})\ell^{(\ell)}(\beta^{(\ell)*}|\Theta^{(\ell)}\backslash\beta^{(\ell)*})\beta^{(\ell)*}}{P(\beta^{(\ell)})\ell^{(\ell)}(\beta^{(\ell)}|\Theta^{(\ell)}\backslash\beta^{(\ell)})\beta^{(\ell)}}$$

such that $\beta^{(\ell)*}$ is proposed parameter.

2.3.4 update $u_i^{(\ell)}$

with likelihood function same as above, and using hierarchical prior with symmetric proposal distribution(in this case, normal distribution) we can defince MH as follow.

$$\text{MH ratio} = \frac{P(u_i^{(\ell)*}|z_i)\ell^{(\ell)}(u_i^{(\ell)*}|\Theta^{(\ell)}\backslash u_i^{(\ell)*})}{P(u_i^{(\ell)}|z_i)\ell^{(\ell)}(u_i^{(\ell)}|\Theta^{(\ell)}\backslash u_i^{(\ell)})}$$

such that $u_i^{(\ell)*}$ is proposed parameter.

2.3.5 update $z_i^{(\ell)}$

global latent positions only depend on positions of each layer's latent position. therefore we can define likelihood function as follows.

$$\ell(z_i|\boldsymbol{u},\boldsymbol{\sigma}) = \prod_{\ell\in L}\prod_{i<j} P(u_i^{(\ell)}|z_i,\sigma^2_{(\ell)})$$

such that $\boldsymbol{u} = \{u_1^{(1)}, ...u_i^{(\ell)}, ...\}, \boldsymbol{\sigma} = \{\sigma_1, ..., \sigma_\ell, ...\}$ therefore $\boldsymbol{u}, \boldsymbol{\sigma}$ are set of parameters that containing entire latent positions of all layer and corresponding variances. in that sense, we can define MH ratio as follow.

$$\text{MH ratio} = \frac{P(z_i^*)\ell(z_i^*|\boldsymbol{u},\boldsymbol{\sigma})}{P(z_i)\ell(z_i|\boldsymbol{u},\boldsymbol{\sigma})}$$

References

Sosa, Juan. Betancourt, B. (2022). A latent space model for multilayer network data. *Computational Statistics & Data Analysis*.